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RNA-based co-transfer of human *CD8αβ* with WT1-specific *TCRαβ* redirects tumor recognition by CD4 and $\gamma\delta$ T-cells towards MHC class I-restricted WT1 epitopes and boosts CD8 T-cell responses with or without *CD3* mRNA

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Abstract

We evaluated the redirection of CD4, $\gamma\delta$ and CD8 T-cells towards the Wilms' tumor protein (WT1) tumor-associated antigen, using a major histocompatibility complex (MHC) class I-restricted WT1-specific T-cell receptor (TCR) introduced via RNA-based engineering. We also studied whether co-transfection of *TCR* mRNA in combination with *CD8αβ* mRNA in CD4 and $\gamma\delta$ T-cells or with *CD8αβ* and *CD3γδ* mRNAs in CD8 T-cells improves antigen-specific T-cell functional activity. We transfected primary human CD4 and CD8 T-cells following our in-house-developed protocol, in which electroporation with Dicer-substrate silencing RNA (DsiRNA) suppresses de novo expression of native TCR, followed by DsiRNA-resistant transgenic *TCR* mRNA transfection. This method allows minimal mispairing between native and introduced TCR chains. High frequencies of transgenic MHC class I-restricted WT1-specific TCR-positive cells were obtained in expanded CD4 and $\gamma\delta$ T-cells. Only co-electroporation of *CD8* mRNA led *TCR* mRNA-electroporated CD4 and $\gamma\delta$ T-cells to MHC class I-restricted antigen-specific recognition of tumor cells.

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Co-electroporation of CD8 T-cells with WT1-specific *TCR*, *CD8* and *CD3* mRNAs also enhanced CD8 T-cell activation and antigen-specific recognition as compared to either TCR-engineered or TCR- and CD8-engineered cells. In summary, RNA electroporation is a fast and efficient method to engineer primary human CD8, CD4 and $\gamma\delta$ T-cells for redirecting T-cell specificity. Transgenic CD8 expression in CD4 and $\gamma\delta$ T-cells and co-electroporation of *CD8* and *CD3* mRNA in CD8 T-cells enable antigen recognition when T-cells are redirected with TCRs of low/intermediate avidity, showing the potential of TCR co-receptors to improve T-cell functional activity against tumor-associated antigens in adoptive TCR-T-cell therapies.

Keywords T-cell redirection, T-cell receptor, Non-viral engineering, Messenger RNA, T-cell coreceptors

To the Editor:

T-cell receptor (TCR)-mediated cancer therapy has focused on redirecting CD8 T-cells [1]. CD4 [2] and $\gamma\delta$ T-cells [3] have gained attention for adoptive T-cell treatments, but lack the CD8 co-receptor, which otherwise modulates TCR activity by interacting with major histocompatibility complex (MHC) class I molecules [4, 5]. TCRs reacting against cancer-associated self-antigens, e.g. the Wilms' tumor protein (WT1) [6], are of low/intermediate affinity following negative selection in the thymus [4]. We aimed to enhance tumor recognition by intermediate-affinity MHC class I-restricted WT1-specific TCRs, by expressing co-receptors linked to TCR activity, using non-integrative RNA-based electroporation [7] (Additional file 1: Methods). In this study we used mRNA encoding a CD8 T-cell-derived HLA-A*02:01-restricted WT1₃₇₋₄₅-specific TCR (T37), which has an intermediate functional avidity of 19.6 nM, as measured by the half-maximal effective concentration (EC50) against titrated peptide on the U266 cell line, which has a very low intracellular expression of WT1 [5].

Similar to our observations in CD8 T-cells [7], electroporation of Dicer-substrate silencing RNAs (DsiRNAs) directed against TCR alpha (*TRAC*) and beta constant (*TRBC*) regions of wild-type TCR $\alpha\beta$ chains reduced the expression of native TCRs in CD4 T-cells (not shown). Following DsiRNA electroporation, CD4 T-cells were electroporated with the T37 mRNA alone or in combination with *CD8* mRNA, showing a significant increase in T37 *TCR* mRNA levels (not shown). For $\gamma\delta$ T-cells, a single electroporation protocol either with T37 *TCR*, *CD8* or both mRNAs was performed, as $\gamma\delta$ T-cells do not express a native TCR $\alpha\beta$ and do not require a first electroporation of DsiRNAs against *TRAC* and *TRBC* to prevent TCR $\alpha\beta$ mispairing. Surface expression of T37 TCR was detected in $\gamma\delta$ and expanded DsiRNA-pretreated CD4 T-cells (respectively T37 and ST37 conditions); and became significantly detectable by *CD8* mRNA co-electroporation (ST37 + *CD8* condition) in resting and expanded CD4 T-cells (Fig. 1).

T37-engineered expanded CD4 and $\gamma\delta$ T-cells were examined whether they recognize tumor cells in an antigen-specific manner. We compared the upregulation of the CD25, CD69, CD137 and CD154 activation

markers [8] and the down-regulation of CD127, which are seen in CD4 T-cells after antigen engagement [9, 10]. Co-electroporation with *CD8* mRNA in ST37-engineered CD4 T-cells (ST37 + *CD8* condition) resulted in a significant increase of CD25 expression in co-cultures with HLA-A*02-positive WT1₃₇₋₄₅ peptide-pulsed T2 cells and Raji-A2 cells, and of CD69, CD137, and CD154 in co-cultures with WT1₃₇₋₄₅ peptide-pulsed T2 cells (Additional file 2: Figure S1A); in a significant downregulation of CD127 in response to WT1₃₇₋₄₅ peptide-pulsed T2 and Raji-A2 cells (Additional file 2: Figure S1B); and in a significant increase in granzyme B secretion, a surrogate marker for cell-mediated cytotoxicity, in response to WT1₃₇₋₄₅ peptide-pulsed T2 and Raji-A2 cells (Additional file 2: Figure S2). Co-electroporation of *CD8* mRNA significantly increased the ability of T37-engineered $\gamma\delta$ T-cells to specifically kill HLA-A*02-positive WT1₃₇₋₄₅ peptide-pulsed T2, U266, K562-A2 and THP1 tumor cells (Additional file 2: Figure S3). We conclude that *de novo* transgenic expression of CD8 in primary CD8-negative T-cells via *CD8* mRNA electroporation leads to MHC class I-restricted antigen-specific recognition of tumor cells. This is consistent with previous studies showing that co-introducing CD8 increases the T-cell functionality of CD4 [2, 11, 12] and $\gamma\delta$ T-cells [13], transfected with tumor-specific TCR of intermediate affinity.

Peptide-pulsed T2 but not Raji-A2 cells induced CD69, CD137 and CD154 in ST37 + *CD8*-redirected CD4 T-cells and cytotoxicity in T37 + *CD8*-engineered $\gamma\delta$ T-cells (Additional file 2: Figures S1, S3). This is likely due to competition for MHC binding between endogenously processed and exogenous peptides in Raji-A2 but not T2 cells [5, 14]. Activation threshold by Raji-A2 cells is reached less easily due to insufficient WT1 peptide availability.

CD8 mRNA co-electroporation did not succeed in inducing a specific response in CD8-negative T-cells to non-peptide-pulsed tumor cells that intracellularly process and express WT1, e.g. HLA-A*02-positive WT1-transduced Raji cells (Raji-A2-WT1). Therefore, we assessed using our double sequential electroporation protocol [5, 7], whether transfection of DsiRNA- and T37-engineered CD8 T-cells (ST37 condition) with mRNAs encoding CD8 or CD8 and CD3 co-receptors

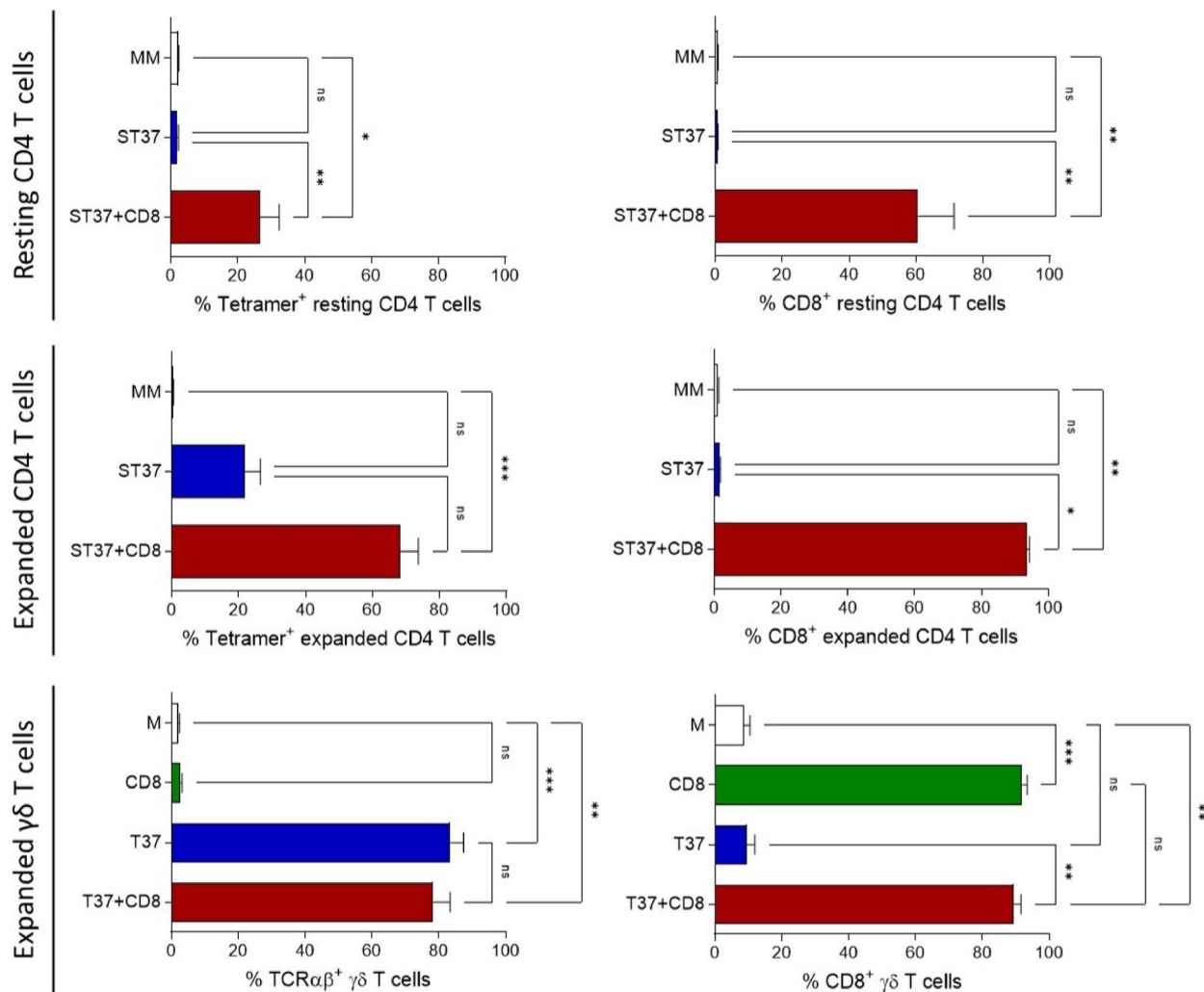


Fig. 1 Transgenic TCR and CD8 expression in mRNA-electroporated primary CD4 and $\gamma\delta$ T-cells. T37 TCR (left panel) and CD8 (right panel) expression was measured 24 h after electroporation with WT1₃₇₋₄₅-specific TCR mRNA with or without CD8 mRNA. Graphs represent mean $\pm$ SD values for 6–8 independent donors that were analyzed using Kruskal-Wallis test followed by corrected Dunn's multiple comparison test (ns, not significant; *, $0.01 < P < 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$). M, control single mock electroporation; MM, control double sequential mock electroporation; ST37, double sequential electroporation, first with DsiRNA directed against *TRAC* and against *TRBC* mRNA, followed by WT1₃₇₋₄₅TCR mRNA electroporation; T37, WT1₃₇₋₄₅TCR mRNA electroporation; +CD8, CD8 mRNA co-electroporated with TCR mRNA; CD8, CD8 mRNA electroporation only

(respectively ST37 + CD8 and ST37 + CD8+CD3 conditions (Additional file 2: Figure S4)) correlated with improved antigen-specific recognition of target cells. T37 TCR expression reached maximum levels in ST37-, ST37 + CD8- and ST37 + CD8+CD3-engineered CD8 T-cells, without significant differences among them (respectively $92.3 \pm 5.6\%$, $88.9 \pm 8.8\%$ and $91.8 \pm 4.7\%$ (mean $\pm$ standard deviation)). Figure 2 shows that in the ST37 + CD8 condition, electroporation with CD8 mRNA significantly increased antigen recognition by ST37-engineered CD8 T-cells on peptide-pulsed U266 cells (increased CD137 expression), on peptide-pulsed Raji-A2 cells (increased CD137 expression and granzyme secretion) and on Raji-A2-WT1 cells (increased CD137 expression). In the ST37 + CD8+CD3 condition, co-electroporation of CD3 with

CD8 mRNA showed an additional significant advantage over electroporation with CD8 mRNA on peptide-pulsed Raji-A2 and on Raji-A2-WT1 cells (increased CD137 expression). The ST37 + CD8+CD3 condition also showed a significant advantage not seen with the ST37 + CD8 condition on unpulsed U266 (increased CD137 expression) and on Raji-A2-WT1 cells (increased granzyme secretion). We conclude that co-electroporation of CD8 and CD3 mRNA in TCR-redirection CD8 T-cells improves antigen-specific tumor recognition, being able to detect and respond to intracellularly processed WT1 epitope.

In conclusion, we provide a non-viral RNA-based protocol to increase the in vitro functional activity of TCR-redirection T-cells by upregulation of TCR co-receptors,

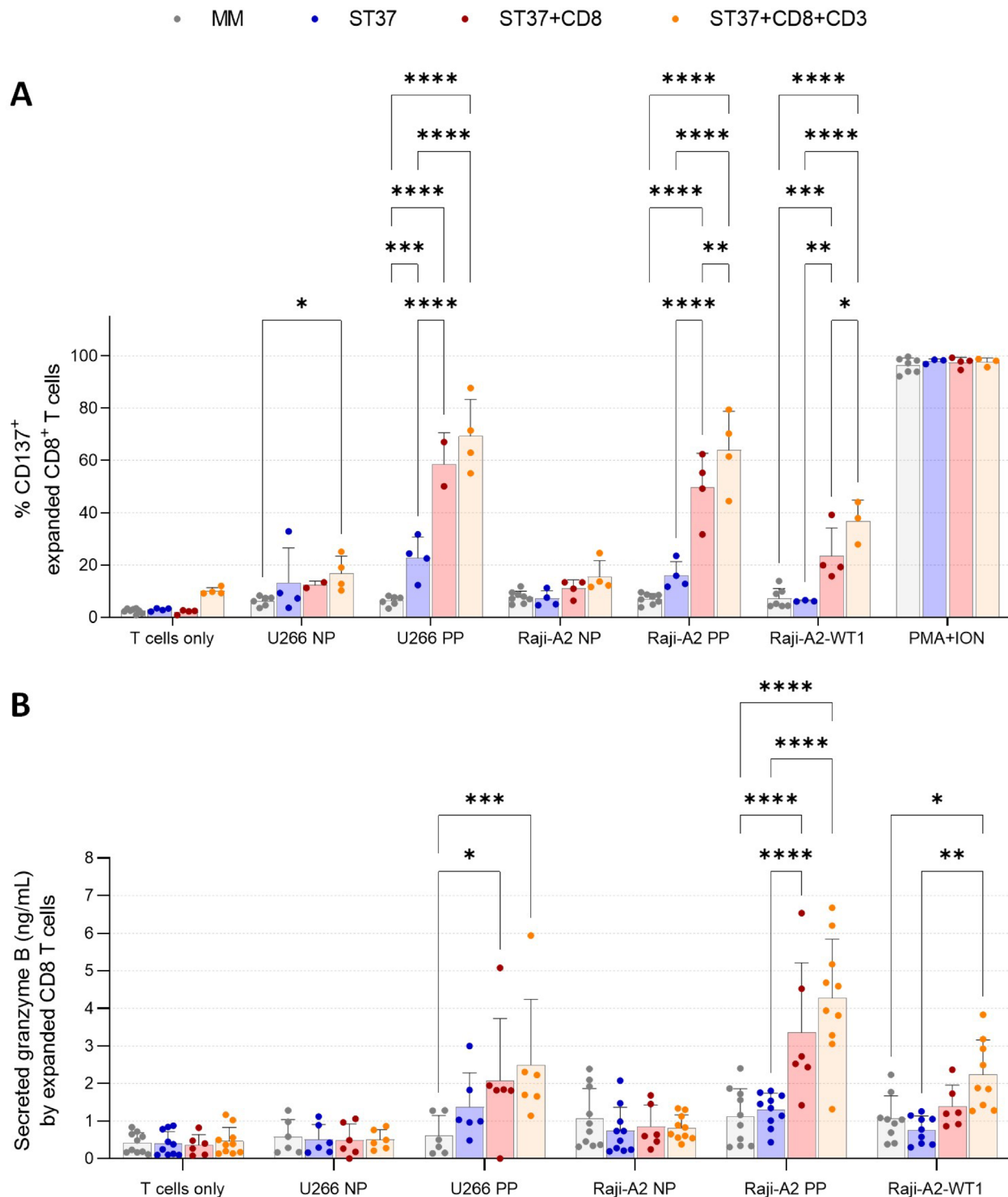


Fig. 2 Antigen-specific functional activity in expanded primary T37 TCR-redirection CD8 T-cells in response to WT1-specific recognition. **A** Upregulation of CD137 activation marker was analyzed 24 h after start of co-cultures of T37 TCR-redirection T-cells and tumor cell lines that were pulsed with WT1₃₇₋₄₅ peptide (PP) or left unpulsed (NP). **B** Secretion of granzyme B was analyzed in supernatants from the same co-cultures using ELISA. As a negative control, T-cells were cultured in the absence of target cells ('T cells only' condition). As a positive control, T-cells were cultured in the presence of phorbol 12-myristate 13-acetate and ionomycin (PMA+ION). CD8 T-cells were expanded for 1 week before RNA electroporation. Graphs represent mean $\pm$ SD values for 3–10 independent donors and were analyzed using two-way ANOVA followed by Tukey's multiple comparison test (only statistically significant differences between groups are shown; *, $0.01 < P < 0.05$; **, $P \leq 0.01$; ***, $P \leq 0.001$; ****, $P \leq 0.0001$). MM, double sequential mock electroporation; ST37, double sequential electroporation, first with DsiRNA directed against *TRAC* and against *TRBC* mRNA, followed by *WT1*₃₇₋₄₅TCR mRNA electroporation; +CD8, *CD8* mRNA co-electroporated with *TCR* mRNA; +CD3, *CD3* mRNA co-electroporated with *TCR* mRNA and *CD8* mRNA

highlighting their importance in the context of TCR-T-cell therapies. While it would have to be confirmed in vivo, this would have the advantage of using transiently expressed TCRs derived from natural repertoires against tumor self-antigens, in order to try and avoid unpredictable treatment-related toxicities, that have been described when using lentivirally integrated tumor-targeting affinity-enhanced TCR sequences [15].

There was WT1₃₇₋₄₅/HLA-A*02:01 tetramer staining in ST37+CD8-electroporated cells in both resting and expanded CD4 T-cells ($26.7 \pm 15.1\%$ and $68.4 \pm 13.2\%$, respectively). In ST37 conditions, T37 TCR was only detected in expanded CD4 T-cells ($21.9 \pm 11\%$). CD8 expression was confirmed in ST37+CD8-electroporated resting ($60.7 \pm 30.2\%$) and expanded ($93.5 \pm 2.1\%$) CD4 T-cells, whereas no CD8 expression was detected in ST37 conditions.

An anti-human pan-TCR $\alpha\beta$ antibody was used to analyze T37 TCR surface levels on $\gamma\delta$ T-cells. T37 TCR was expressed at high similar levels in T37 ($83.5 \pm 11.2\%$) and T37+CD8-electroporated ($78.1 \pm 14.8\%$) $\gamma\delta$ T-cells. A small population of $\gamma\delta$ T cells expressed CD8 (mock: $8.6 \pm 5.2\%$; T37: $9.6 \pm 6.7\%$), but as seen in CD4 T-cells, CD8 levels were greatly enhanced by CD8 mRNA transfection (CD8: $91.9 \pm 4.2\%$; T37+CD8: $89.4 \pm 6.1\%$).

In terms of antigen-specific functionality, in both ST37+CD8 and ST37+CD8+CD3 conditions, CD8 T-cells significantly upregulated CD137 expression upon encounter of WT1₃₇₋₄₅ peptide-pulsed and WT1-expressing tumor cells compared to mock-electroporated and ST37-engineered CD8 T-cells (Fig. 2A). In co-cultures with WT1-expressing Raji-A2-WT1 tumor cells, ST37+CD8+CD3-engineered CD8 T-cells showed a significantly higher upregulation of CD137 ($36.6 \pm 8.2\%$; Fig. 2A), compared to their ST37+CD8 counterpart ($23.5 \pm 10.6\%$). In co-cultures with WT1₃₇₋₄₅ peptide-pulsed Raji-A2 cells, a significantly increased granzyme B secretion was observed in both ST37+CD8- and ST37+CD8+CD3-engineered CD8 T-cells versus mock-electroporated and ST37-engineered CD8 T-cells (Fig. 2B). Importantly, against WT1-expressing Raji-A2-WT1 tumor cells, ST37+CD8+CD3-engineered CD8 T-cells (2.2 ± 0.9 ng/mL), but not those redirected with ST37+CD8 (1.4 ± 0.6 ng/mL), secreted significantly higher amounts of granzyme B in comparison with their ST37 CD8 T-cells counterpart (0.8 ± 0.4 ng/mL; Fig. 2B).

Abbreviations

DsiRNA Dicer	substrate silencing RNA
EC50 Half	maximal effective concentration
HLA	Human leukocyte antigen
ION	ionomycin
M	Singlemock electroporation
MHC	Major histocompatibility complex
MM	Double sequential mock electroporation
mRNA	messenger RNA

NP	Non-pulsed (unpulsed) with WT1 ₃₇₋₄₅ peptide
ns	Not significant
PMA	Phorbol 12-myristate 13-acetate
PP	Peptide-pulsed (pulsed with WT1 ₃₇₋₄₅ peptide)
RNA	Ribonucleic acid
SD	Standard deviation
ST37	Double sequential electroporation, first with DsiRNA directed against <i>TRAC</i> and against <i>TRBC</i> mRNA, followed by WT1 ₃₇₋₄₅ TCR mRNA electroporation
ST37+CD8	Double sequential electroporation, first with DsiRNA directed against <i>TRAC</i> and against <i>TRBC</i> mRNA, followed by WT1 ₃₇₋₄₅ TCR mRNA + CD8 mRNA co-electroporation
ST37+CD8+CD3	Double sequential electroporation, first with DsiRNA directed against <i>TRAC</i> and against <i>TRBC</i> mRNA, followed by WT1 ₃₇₋₄₅ TCR mRNA + CD8 mRNA + CD3 mRNA co-electroporation
TCR	T-cell receptor
TRAC	TCR alpha constant region
TRBC	TCR beta constant region
T37	WT1 ₃₇₋₄₅ TCR mRNA (electroporation)
T37+CD8	WT1 ₃₇₋₄₅ TCR mRNA and CD8 mRNA co-electroporation
WT1	Wilms'tumor protein

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40164-026-00764-6>.

Supplementary Material 1

Supplementary Material 2

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DCD, ZNB, FF, VFIVT, HS and EL conceptualized the research; FF and HS isolated and cloned *TCR* and *CD8* sequences and provided helpful suggestions during repeated scientific discussions; DCD, DF, GR, MV and HHVA performed experiments; DCD and ZNB wrote the manuscript; DF, VFIVT and EL edited the manuscript; ZNB, VFIVT, SA and EL supervised the project. All authors have agreed to this manuscript.

Author contributions

DCD, ZNB, FF, VFIVT, HS and EL conceptualized the research; FF and HS isolated and cloned *TCR* and *CD8* sequences and provided helpful suggestions during repeated scientific discussions; DCD, DF, GR, MV and HHVA performed experiments; DCD and ZNB wrote the manuscript; DF, VFIVT and EL edited the manuscript; ZNB, VFIVT, SA and EL supervised the project. All authors have agreed to this manuscript.

Data availability

The original contributions presented in the study are included in the correspondence article and additional files. Further inquiries can be directed to the corresponding authors.

Declarations

Competing interests

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest. VFIVT and ZNB are coinventors of a now-elapsed patent covering the messenger RNA electroporation technique (Improved Transfection of Eukaryotic Cells with Linear Polynucleotides by Electroporation, WO/2003/000907). ZNB was a consultant for BioNTech.

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